



Submitted by,

Course Title: Operating Systems

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Lab Report: 01

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CSE 2305: Assignment 01

Simulate the Round-Robin (RR) CPU scheduling algorithm for the given data in Table 1. Using the simulator, analyze how different values of the time quantum affect scheduling performance.

Table 1: Process Data

PID	Arrival time	CPU burst
P1	0	10
P2	2	4
P3	4	6
P4	6	8
P5	8	2

Tasks

- Implement RR (handle processes arriving at different times).
- Run for time_quantum {1,2,4,8}.
- Short written analysis (3–6 bullet points) of observed trade-offs and recommendations for choosing quantum.

Table 1: Process data

PID	Arrival time	CPU burst
P1	0	16
P2	2	4
P3	4	6
P4	6	8
P5	8	2

Task:

a) Implementation Round Robin Works:

1. Pick the first process from the front ready queue.

2. Assign the CPU and set the time quantum.

3. Execute the process.

4. If the process finished before TQ ends:

- i) Process terminates
- ii) Scheduler select the next process.

5. The process does not finished before TQ ends:

i) The process is moved to the end of the ready queue

ii) Scheduler selects the next process.

iii) Timer interrupt occurs

iv) Process moves to the back of ready queue

v) Scheduler selects the next process.

6. Repeat this until all processes are completed.

Task B :

AT = Arrival time

CT = Completion time

BT = Burst time

TAT = Turn Around time

RT = Response time

WT = Waiting time

We know that

$$TAT = CT - AT \text{ and}$$

$$WT = TAT - BT$$

RT = process getting the CPU for the first time - AT

RR for TQ = 1

Process	AT	BT	CT	TAT	WT	RT
P1	0	880,7	28	28	18	0
P2	2	42 0	13	11	7	0
P3	4	216 84,3	25	21	15	1
P4	6	482 8,7	30	24	16	2
P5	8	210	17	9	7	3

Queue:

P1	P1	P2	P4	P3	P1	P2	P4	P3	P1	P5	P2	P4	P3
P1	P5	P4	P3	P1	P4	P3	P1	P4	P3	P4	P1	P4	P3

Grantt chart

P1	P1	P2	P1	P2	P3	P1	P2	P4	P3	P1	P5	P2	P4	P3	P1
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

P5	P4	P3	P1	P4	P3	P1	P3	P1	P4	P1	P4	P4
17	18	19	20	21	22	23	24	25	26	27	28	29

$$\text{Average TAT} = ((25 + 11 + 21 + 24 + 9) / 5)$$

$$= (90 / 5) = 18.6$$

$$\text{Average WT} = ((18 + 7 + 15 + 16 + 7) / 5)$$

$$= (63 / 5) = 12.6$$

$$\text{Average RT} = ((0 + 0 + 1 + 2 + 3) / 5)$$

$$= (6 / 5) = 1.2$$

RR for TQ = 2

Process	AT	BT	CT	TAT	WT	RT
P1	0	28 46	28	28	18	0
P2	2	8 0	10	8	4	0
P3	4	6 0	24	20	14	2
P4	6	8 40	20	24	16	4
P5	8	20	16	8	6	6

Queue : (2 | 8) =

P1	P2	P1	P3	P2	P4	P1	P5	P3	P4	P1	P3	P4	P1	P4
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$$\text{Average TAT} = (28 + 8 + 20 + 24 + 8) / 5$$

$$= 17.6$$

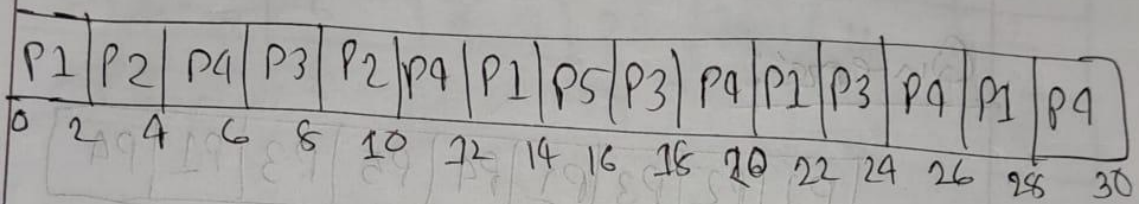
$$\text{Average WT} = (18 + 4 + 14 + 16 + 6) / 5$$

$$= 11.6$$

$$\text{Average RT} = (0+0+2+4+6)/5$$

$$= 2.4$$

Gantt Chart: 2.8 =



RR for TQ = 4

Process	AT	BT	CT	TAT	WT	RT
P1	0	16 0	26	26	16	0
P2	2	4 0	8	6	2	2
P3	4	16 0	24	20	14	4
P4	6	8 0	30	24	16	10
P5	8	2 0	22	14	12	12

$$\text{Average TAT} = 90/5$$

$$= 18$$

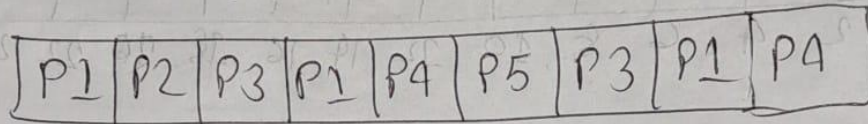
$$\text{Average WT} = 60 / 5$$

$$= 12$$

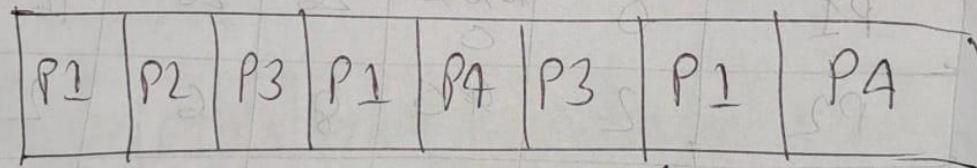
$$\text{Average RT} = 28 / 5$$

$$= 5.6$$

Queue:



Gantt chart:

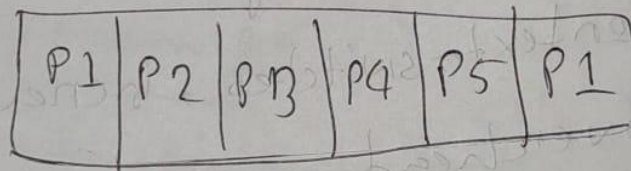


0 4 8 16 20 22 24 26 30

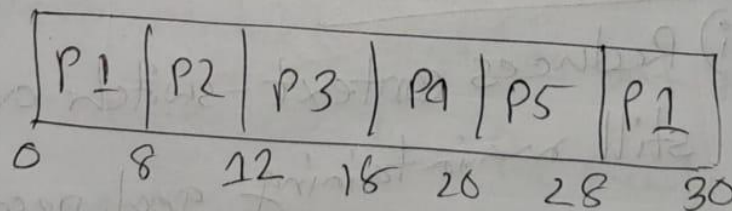
RR For TQ = 8

Process	AT	BT	CT	TAT	WT	RT
P1	0	10 20	36	36	20	0
P2	2	4 0	12	10	6	6
P3	4	8 0	18	14	8	8
P4	6	8 0	26	20	12	12
P5	8	20 0	28	20	18	18

Ready Queue



Gantt Chart



$$\begin{aligned} \text{Average TAT} &= (94/5) = 18.8 \\ \text{" WT} &= 64/5 = 12.8 \\ \text{" RT} &= 44/5 = 8.8 \end{aligned}$$

Task C: (short written Analysis)

Time Quantum 1

- ① Provides the fastest response time and ~~excellent~~ excellent fairness because each process gets CPU time quickly.
- ② It causes highest number of context-switches, increasing CPU overhead.

Time Quantum 2

- i) Reduces context-switch overhead, still maintaining good responsiveness.
- ii) Offers a better balance between fairness and CPU efficiency.

Time Quantum 4:

- i) Further decreases the number of context switches and CPU utilization.
- (ii) Provide a good compromise between response time and overall system performance

Time Quantum 8:

- i) Behaves similarly to the FCFS scheduling algorithm
- (ii) Fewest context switches but results in longer response time for newly arrived processes, making it less suitable for interactive system.

Recommendation:

A medium time quantum (around 2-4 units) is generally the best choice because it balance responsiveness, fairness and CPU efficiency.

Very small quantum increase overhead while very large quantum reduce the benefits of Round Robin and make the scheduler behave more like FCFS.